Chapter 18, Problem 14P

Problem

Design a problem to help other students better understand finding the Fourier transform of a variety of time varying functions (do at least three).

Step-by-step solution

Step 1 of 9

(a) Consider

$f(t) = e^{-at}\sin(At+\alpha)u(t)$

Where $a, A,$ and α are constants.

The function can be taken as

$\sin(At+\alpha) = \sin(At)\cos\alpha + \cos(At)\sin\alpha$

$f(t) = \left[\sin(At)\cos\alpha + \cos(At)\sin\alpha\right]e^{-at}u(t)$

Step 2 of 9

$f(t) = e^{-at}\sin(At)\cos\alpha u(t) + e^{-at}\cos(At)\sin\alpha u(t)$

Apply Fourier transform on both sides, we get:

$\mathcal{F}\{f(t)\} = \mathcal{F}\{e^{-at}\sin(At)\cos\alpha u(t) + e^{-at}\cos(At)\sin\alpha u(t)\}$

$F(\omega) = \cos\alpha\mathcal{F}\{e^{-at}\sin(At)u(t)\} + \sin\alpha\mathcal{F}\{e^{-at}\cos(At)u(t)\}$

$\therefore \boxed{F(\omega) = \cos\alpha\left[\frac{A}{(a+j\omega)^2+A^2}\right] + \sin\alpha\left[\frac{a+j\omega}{(a+j\omega)^2+A^2}\right]}$

Step 3 of 9

(b) Consider

$g(t) = \cos(kt)\{u(t+\beta) - u(t-\beta)\}$

$g(t) = \cos(kt)u(t+\beta) - \cos(kt)u(t-\beta)$

$G(\omega) = \int_{-\infty}^{\infty} g(t)e^{-j\omega t}dt$

$G(\omega) = \int_{-\infty}^{\infty} \{\cos(kt)u(t+\beta) - \cos(kt)u(t-\beta)\}e^{-j\omega t}dt$

Step 4 of 9

$$G(\omega)=\int_{-\beta}^{\infty}\cos(kt)e^{-j\omega t}dt-\int_{\beta}^{\infty}\cos(kt)e^{-j\omega t}dt$$

We know that from Euler's identity,

$$\cos(kt)=\frac{e^{jkt}+e^{-jkt}}{2}$$

$$G(\omega)=\int_{-\beta}^{\infty}\left(\frac{e^{jkt}+e^{-jkt}}{2}\right)e^{-j\omega t}dt-\int_{\beta}^{\infty}\left(\frac{e^{jkt}+e^{-jkt}}{2}\right)e^{-j\omega t}dt$$

$$G(\omega)=\frac{1}{2}\int_{-\beta}^{\infty}\left(e^{-j(\omega-k)t}+e^{-j(\omega+k)t}\right)dt-\frac{1}{2}\int_{\beta}^{\infty}\left(e^{-j(\omega-k)t}+e^{-j(\omega+k)t}\right)dt$$

Step 5 of 9

$$G(\omega)=\frac{1}{2}\left\{\int_{-\beta}^{\infty}e^{-j(\omega-k)t}dt+\int_{-\beta}^{\infty}e^{-j(\omega+k)t}dt-\int_{\beta}^{\infty}e^{-j(\omega-k)t}dt-\int_{\beta}^{\infty}e^{-j(\omega+k)t}dt\right\}$$

$$G(\omega)=\frac{1}{2}\left\{\left[\frac{e^{-j(\omega-k)t}}{-j(\omega-k)}\right]_{-\beta}^{\infty}+\left[\frac{e^{-j(\omega+k)t}}{-j(\omega+k)}\right]_{-\beta}^{\infty}-\left[\frac{e^{-j(\omega-k)t}}{-j(\omega-k)}\right]_{\beta}^{\infty}-\left[\frac{e^{-j(\omega+k)t}}{-j(\omega+k)}\right]_{\beta}^{\infty}\right\}$$

$$G(\omega)=\frac{1}{2}\left\{\left[\frac{e^{-\infty}-e^{j(\omega-k)\beta}}{-j(\omega-k)}\right]+\left[\frac{e^{-\infty}-e^{j(\omega+k)\beta}}{-j(\omega+k)}\right]-\left[\frac{e^{-\infty}-e^{-j(\omega-k)\beta}}{-j(\omega-k)}\right]-\left[\frac{e^{-\infty}-e^{-j(\omega+k)\beta}}{-j(\omega+k)}\right]\right\}$$

$$G(\omega)=\frac{1}{2}\left\{\left[\frac{0-e^{j(\omega-k)\beta}}{-j(\omega-k)}\right]+\left[\frac{0-e^{j(\omega+k)\beta}}{-j(\omega+k)}\right]-\left[\frac{0-e^{-j(\omega-k)\beta}}{-j(\omega-k)}\right]-\left[\frac{0-e^{-j(\omega+k)\beta}}{-j(\omega+k)}\right]\right\}$$

Step 6 of 9

$$G(\omega)=\frac{1}{2}\left\{\frac{e^{j(\omega-k)\beta}-e^{-j(\omega-k)\beta}}{j(\omega-k)}+\frac{e^{j(\omega+k)\beta}-e^{-j(\omega+k)\beta}}{j(\omega+k)}\right\}$$

$$G(\omega)=\left\{\frac{1}{(\omega-k)}\frac{e^{j(\omega-k)\beta}-e^{-j(\omega-k)\beta}}{j2}+\frac{1}{(\omega+k)}\frac{e^{j(\omega+k)\beta}-e^{-j(\omega+k)\beta}}{j2}\right\}$$

$$G(\omega)=\left\{\frac{1}{(\omega-k)}\sin\left[\beta(\omega-k)\right]+\frac{1}{(\omega+k)}\sin\left[\beta(\omega+k)\right]\right\}$$

$$\therefore\boxed{G(\omega)=\frac{\sin\left[\beta(\omega-k)\right]}{\omega-k}+\frac{\sin\left[\beta(\omega+k)\right]}{\omega+k}}$$

Step 7 of 9

(c) Consider

$$h(t)=e^{bt}\sin(pt)u(-t)$$

Let $x(t)=e^{bt}\sin(-pt)u(-t)$

$$x(t)=-e^{bt}\sin(pt)u(-t)$$

And $y(t)=e^{-bt}\sin(pt)u(t)$

Then $x(t)=y(-t)$

And $h(t)=-x(t)$

Step 8 of 9

Apply Fourier transform on both sides of $y(t)$, we get:

$$Y(\omega)=\frac{p}{(b+j\omega)^2+p^2}$$

$$X(\omega)=Y(-\omega)$$

$$X(\omega)=\frac{p}{(b-j\omega)^2+p^2}$$

Step 9 of 9

Hence

$$H(\omega)=-X(\omega)$$

$$H(\omega)=-\left[\frac{p}{(b-j\omega)^2+p^2}\right]$$

$$\therefore \boxed{H(\omega)=\frac{-p}{(b-j\omega)^2+p^2}}$$